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ROVANAEST
Ropivacaine Hydrochloride Solution for Injection or Infusion USP

Composition:
ROVANAEST 2.0 (Ropivacaine Hydrochloride 2 mg/mL Solution for Injection or Infusion USP):
1 mL of ROVANAEST 2.0 contains 2.116 mg Ropivacaine Hydrochloride Monohydrate equivalent to 2 mg Ropivacaine Hydrochloride.
20 mL ampoule contains 42.3 mg Ropivacaine Hydrochloride Monohydrate.

ROVANAEST 7.5 (Ropivacaine Hydrochloride 7.5 mg/mL Solution for Injection or Infusion USP)
1 mL of ROVANAEST 7.5 contains 7.93 mg Ropivacaine Hydrochloride Monohydrate equivalent to 7.5 mg Ropivacaine Hydrochloride.
20 mL ampoule contains 158.6 mg Ropivacaine Hydrochloride Monohydrate.

Product Description:
ROVANAEST is a sterile, clear and colourless solution.

Pharmacodynamics:
Mechanism of action
Ropivacaine is a long-acting, amide-type local anaesthetic with both anaesthetic and analgesic effects. At high doses, ROVANAEST produces surgical anaesthesia, while at lower doses it produces sensory block with limited and non-progressive motor block.

The mechanism is a reversible reduction of the membrane permeability of the nerve fibre to sodium ions. Consequently, the depolarisation velocity is decreased and the excitable threshold increased, resulting in a local blockade of nerve impulses.

The most characteristic property of Ropivacaine is the long duration of action. Onset and duration of the local anaesthetic efficacy are dependent upon the administration site and dose, but are not influenced by the presence of a vasoconstrictor (e.g. adrenaline (epinephrine)).

Healthy volunteers exposed to intravenous infusions tolerated Ropivacaine well at low doses and with expected CNS symptoms at the maximum tolerated dose. The clinical experience with this drug indicates a good margin of safety when adequately used in recommended doses.

Pharmacokinetics:
Absorption
Ropivacaine has a chiral center and is available as the pure S-(-)-enantiomer. It is highly lipid-soluble. All metabolites have a local anaesthetic effect but of considerably lower potency and shorter duration than that of Ropivacaine.

There is no evidence of in vivo racemisation of Ropivacaine.

The plasma concentration of Ropivacaine depends upon the dose, the route of administration and the vascularity of the injection site. Ropivacaine follows linear pharmacokinetics and the C_{max} is proportional to the dose.

Ropivacaine shows complete and biphasic absorption from the epidural space with half-lives of the two phases of the order of 14 minutes and 4 hours in adults. The slow absorption is the rate-limiting factor in the elimination of ropivacaine, which explains why the apparent elimination half-life is longer after epidural than after intravenous administration. Ropivacaine shows a biphasic absorption from the caudal epidural space also in children.

Distribution
Ropivacaine has a mean total plasma clearance in the order of 440 mL/min, a renal clearance of 1 mL/min, a volume of distribution at steady state of 47 litres and a terminal half-life of 1.8 h after intravenous administration. Ropivacaine has an intermediate hepatic extraction ratio of about 0.4. It is mainly bound to α1-acid glycoprotein in plasma with an unbound fraction of about 6%.

An increase in total plasma concentrations during continuous epidural and interscalene infusion has been observed, related to a postoperative increase of α1-acid glycoprotein.

Variations in unbound, i.e. pharmacologically active, concentration have been much less than in total plasma concentration.

Elimination
Since Ropivacaine has an intermediate to low hepatic extraction ratio, its rate of elimination should depend on the unbound plasma concentration. A postoperative increase in AAG will decrease the unbound fraction due to increased protein binding, which will decrease the total clearance and result in an increase in total plasma concentrations, as seen in the paediatric and adult studies. The unbound clearance of Ropivacaine remains unchanged as illustrated by the stable unbound concentrations during postoperative infusion. It is the unbound plasma concentration that is related to systemic pharmacodynamic effects and toxicity.

Ropivacaine readily crosses the placenta and equilibrium in regard to unbound concentration will be rapidly reached. The degree of plasma protein binding in the foetus is less than in the mother, which results in lower total plasma concentrations in the foetus than in the mother.

Ropivacaine is extensively metabolised, predominantly by aromatic hydroxylation. In total, 86% of the dose is excreted in the urine after intravenous administration, of which only about 1% relates to unchanged drug. The major metabolite is 3-hydroxy-ropivacaine, about 37% of which is excreted in the urine, mainly conjugated. Urinary excretion of 4-hydroxy-ropivacaine, the N-dealkylated metabolite (PPX) and the 4-hydroxy-dealkylated accounts for 1–3%. Conjugated and unconjugated 3-hydroxy-ropivacaine shows only detectable concentrations in plasma.

A similar pattern of metabolites has been found in children above one year.

Impaired renal function has little or no influence on Ropivacaine pharmacokinetics. The renal clearance of PPX is significantly correlated with creatinine clearance. A lack of correlation between total exposure, expressed as AUC, with creatinine clearance indicates that the total clearance of PPX includes a non-renal elimination in addition to renal excretion. Some patients with impaired renal function may show an increased exposure to PPX resulting from a low non-renal clearance. Due to the reduced CNS toxicity of PPX as compared to ropivacaine the clinical consequences are considered negligible in short-term treatment. Patients with end-stage renal disease undergoing dialysis have not been studied.

Paediatrics
The pharmacokinetics of ropivacaine was characterized in a pooled population PK analysis on data in 192 children between 0 and 12 years. Unbound Ropivacaine and PPX clearance and ropivacaine unbound volume of distribution depend on both body weight and age up to the maturity of liver function, after which they depend largely on body weight. The maturation of unbound ropivacaine clearance appears to be complete by the age of 3 years, that of PPX by the age of 1 year and unbound Ropivacaine volume of distribution by the age of 2 years. The PPX unbound volume of distribution only depends on body weight. As PPX has a longer half-life and a lower clearance, it may accumulate during epidural infusion while unbound Ropivacaine clearance (Cl_u) for ages above 6 months has reached values within the range of those in adults.

Indication:
ROVANAEST 7.5:
Surgical anaesthesia
- Epidural block for surgery, including Caesarean section
- Major nerve block
- Field block

Acute pain management
- Continuous epidural infusion or intermittent bolus administration e.g. postoperative or labour pain
- Field block

ROVANAEST 2.0:
Acute pain relief (peri- and postoperative) in children:
- Caudal epidural block in infants (>30 days old) and children up to and including 12 years.
- Continuous epidural infusion in infants (>30 days old) and children up to and including 12 years.

Recommended Dosage:
ROVANAEST should only be used by, or under the supervision of, clinicians experienced in regional anaesthesia.

Posology
ADULTS AND ADOLESCENTS ABOVE 12 YEARS OF AGE:
The following table is a guide to dosage for the more commonly used blocks. The clinician’s experience and knowledge of the patient’s physical status are of importance when deciding the dose. In general, surgical anaesthesia (e.g. epidural administration) requires the use of the higher concentrations and doses. For analgesia the 2 mg/mL concentration of ROVANAEST is generally recommended.

Table 1: Dosage and recommendations for ROVANAEST in adults and children above 12 years of age

	Conc. mg/mL	Volume mL	Dose mg	Onset minutes	Duration hours	
SURGICAL ANAESTHESIA						
Lumbar Epidural Administration						
Surgery	7.5	15–25	113–188	10–20	3–5	The doses in the table are those considered to be necessary to produce a successful block and should be regarded as guidelines for use in adults. Individual variations in onset and duration occur. The figures in the column ‘Dose’ reflect the expected average dose range needed. Standard textbooks should be consulted for both factors affecting specific block techniques and individual patient requirements.
Caesarean section	7.5	15–20	113–150 ⁽¹⁾	10–20	3–5	
Thoracic Epidural Administration						
To establish block for postoperative pain relief	7.5	5–15 (depending on the level of injection)	38–113	10–20	n/a ⁽²⁾	⁽¹⁾ Incremental dosing should be applied, the starting dose of about 100 mg (97.5 mg = 13 mL; 105 mg = 14 mL) to be given over 3-5 minutes. Two additional doses, each of 25 mg, may be administered as needed. The total administered dose should not exceed 150 mg.
Major Nerve Block (e.g. brachial plexus)	7.5	30–40	225–300	10–25	6–10	
Field Block (e.g. minor nerve blocks and infiltration)	7.5	1–30	7.5–225	1–15	2–6	
ACUTE PAIN MANAGEMENT						
Lumbar Epidural Administration						
Bolus	2.0	10–20	20–40	10–15	0.5–1.5	⁽²⁾ n/a = not applicable.
Intermittent injections (top up) (e.g. labour pain management)	2.0	10–15 (minimum interval 30 mins)	20–30			
Continuous infusion e.g. labour pain	2.0	6–10 mL/h	12–20 mg/h	n/a ⁽²⁾	n/a ⁽²⁾	
Postoperative pain management	2.0	6–14 mL/h	12–28 mg/h	n/a ⁽²⁾	n/a ⁽²⁾	⁽³⁾ The dose for a major nerve block must be adjusted according to site of administration and patient status. Supraclavicular brachial plexus blocks may be associated with a higher frequency of serious adverse reactions, regardless of the local anaesthetic used.
Thoracic Epidural Administration						
Continuous infusion (postoperative pain management)	2.0	6–14 mL/h	12–28 mg/h	n/a ⁽²⁾	n/a ⁽²⁾	
Field Block (e.g. minor nerve blocks and infiltration)	2.0	1–100	2–200	1-5	2–6	

Method of administration
In order to avoid intravascular injection, aspiration should be repeated prior to and during administration of the main dose, which should be injected slowly or in incremental doses, at a rate of 25-30 mg/min, while closely observing the patient’s vital functions and maintaining verbal contact. When an epidural dose is to be injected, a preceding test dose of 3-5 mL Lidocaine (Lignocaine) with Adrenaline (Xylocaine 1-2% with Adrenaline 1:200,000) is recommended. An inadvertent intravascular injection may be recognised by a temporary increase in heart rate and an accidental intrathecal injection by signs of a spinal block.

If toxic symptoms occur, the injection should be stopped immediately.

In epidural block for surgery, single doses of up to 250 mg Ropivacaine have been used and well tolerated.

When prolonged blocks are used, either through continuous infusion or through repeated bolus administration, the risks of reaching a toxic plasma concentration or inducing local neural injury must be considered. Cumulative doses up to 675 mg Ropivacaine for surgery and postoperative analgesia administered over 24 hours were well tolerated in adults, as were postoperative continuous epidural infusions at rates up to 28 mg/hour for 72 hours. In a limited number of patients, higher doses of up to 800 mg/day have been administered with relatively few adverse reactions.

For treatment of postoperative pain, the following technique can be recommended: Unless preoperatively instituted, an epidural block with ROVANAEST 7.5 is induced via an epidural catheter. Analgesia is maintained with ROVANAEST 2.0. Infusion rates of 6-14 mL (12-28 mg) per hour provide adequate analgesia with only slight and non-progressive motor block in most cases of moderate to severe postoperative pain. The maximum duration of epidural block is 3 days. However, close monitoring of analgesic effect should be performed in order to remove the catheter as soon as the pain condition allows it. With this technique a significant reduction in the need for opioids has been observed.

In clinical studies an epidural infusion of ROVANAEST 2.0 alone or mixed with fentanyl 1-4 microgram/mL has been given for postoperative pain management for up to 72 hours. ROVANAEST 2.0 (6-14 mL/hour) provided adequate pain relief for the majority of patients. The combination of ROVANAEST and fentanyl provided improved pain relief but caused opioid side effects. The combination of ROVANAEST and fentanyl has been investigated only for ROVANAEST 2.0.

For Caesarean section, neither intrathecal administration nor the use of the Ropivacaine Concentration 10 mg/mL for epidural administration, have been documented.

ROVANAEST 2.0
Paediatric population
Table 2: Children (>30 days old and up to and including 12 years):

	Conc. mg/mL	Volume mL/kg	Dose mg/kg	
ACUTE PAIN MANAGEMENT (peri- and postoperative)				^a Doses in the low end of the dose interval are recommended for thoracic epidural blocks while doses in the high end are recommended for lumbar or caudal epidural blocks.
Single Caudal Epidural Block in children >30 days old and up to 12 years Blocks below T12, in children with a body weight up to 25 kg	2 mg/mL	1 mL/kg	2 mg/kg	
Continuous Epidural Infusion In children with a body weight up to 25 kg >30 days old and up to 6 months				^b Recommended for lumbar epidural blocks.
Bolus dose ^a	2 mg/mL	0.5–1 mL/kg	1–2 mg/kg	
Infusion up to 72 hours 6 up to 12 months	2 mg/mL	0.1 mL/kg/h	0.2 mg/kg/h	The recommended strength of ROVANAEST solution for injection/ infusion for Single Caudal Epidural Block in children >30 days old and up to 12 years and for Continuous Epidural Infusion in children with a body weight up to 25 kg is 2 mg/mL.
Bolus dose ^a	2 mg/mL	0.5–1 mL/kg	1–2 mg/kg	
Infusion up to 72 hours	2 mg/mL	0.2 mL/kg/h	0.4 mg/kg/h	
1 up to 12 years				
Bolus dose ^b	2 mg/mL	1 mL/kg	2 mg/kg	
Infusion up to 72 hours	2 mg/mL	0.2 mL/kg/h	0.4 mg/kg/h	

Method of administration
To prevent inadvertent intravascular injections, great caution should be observed. Careful aspiration is recommended before and during injection of the total dose. The patient’s vital functions should be carefully monitored during the injection. Should toxic signs appear, the injection should be immediately stopped.

When administering the calculated dose, fractionation of the total dose is always recommended.

A single caudal epidural injection of ROVANAEST 2.0 produces adequate postoperative analgesia below T12 in the majority of patients when a dose of 2 mg/kg is used in a volume of 1 mL/kg. The volume of the caudal epidural injection may be adjusted to control the spread of the sensory block. Doses up to 3 mg/kg of a concentration of ropivacaine 3 mg/mL have been used safely in children older than 4 years.

For children with a body weight over 25 kg there is limited experience of caudal blocks. The use of ropivacaine in premature children has not been documented.

Route of administration: For field block and epidural administration only. Not for intravenous infusion.

Contraindications:
Hypersensitivity to Ropivacaine or to other local anaesthetics of the amide type.
General contraindications related to epidural anaesthesia, regardless of the local anaesthetic used, should be taken into account.
Intravenous regional anaesthesia.
Obstetric paracervical anaesthesia.
Hypovolaemia.

Warnings and Precautions:
Regional anaesthetic procedures should always be performed in a properly equipped and staffed area. Equipment and drugs necessary for monitoring and emergency resuscitation should be immediately available. Patients receiving major blocks should be in an optimal condition and have an intravenous line inserted before the blocking procedure. The clinician responsible should take the necessary precautions to avoid intravascular injection and be appropriately trained and familiar with diagnosis and treatment of side effects, systemic toxicity and other complications such as inadvertent subarachnoid injection, which may produce a high spinal block with apnoea and hypotension. Convulsions have occurred most often after brachial plexus block and epidural block. This is likely to be the result of either accidental intravascular injection or rapid absorption from the injection site.

Caution is required to prevent injections in inflamed area.

Cardiovascular
Epidural and intrathecal anaesthesia may lead to hypotension and bradycardia. Hypotension should be treated promptly with a vasopressor intravenously, and with an adequate vascular filling.

Patients treated with anti-arrhythmic drugs class III (e.g. amiodarone) should be under close surveillance and ECG monitoring considered, since cardiac effects may be additive.

There have been rare reports of cardiac arrest during the use of ROVANAEST for epidural anaesthesia or peripheral nerve blockade, especially after unintentional accidental intravascular administration in elderly patients and in patients with concomitant heart disease. In some instances, resuscitation has been difficult. Should cardiac arrest occur, prolonged resuscitative efforts may be required to improve the possibility of a successful outcome.

Head and neck blocks

Certain local anaesthetic procedures, such as injections in the head and neck regions, may be associated with a higher frequency of serious adverse reactions, regardless of the local anaesthetic used.

Major peripheral nerve blocks

Major peripheral nerve blocks may imply the administration of a large volume of local anaesthetic in highly vascularised areas, often close to large vessels where there is an increased risk of intravascular injection and/or rapid systemic absorption, which can lead to high plasma concentrations.

Hypersensitivity

A possible cross–hypersensitivity with other amide–type local anaesthetics should be taken into account.

Hypovolaemia

Patients with hypovolaemia due to any cause can develop sudden and severe hypotension during epidural anaesthesia, regardless of the local anaesthetic used.

Patients in poor general health

Patients in poor general condition due to ageing or other compromising factors such as partial or complete heart conduction block, advanced liver disease or severe renal dysfunction require special attention, although regional anaesthesia is frequently indicated in these patients.

Patients with hepatic and renal impairment

Ropivacaine is metabolised in the liver and should therefore be used with caution in patients with severe liver disease; repeated doses may need to be reduced due to delayed elimination. Normally there is no need to modify the dose inpatients with impaired renal function when used for single dose or short-term treatment. Acidosis and reduced plasma protein concentration, frequently seen in patients with chronic renal failure, may increase the risk of systemic toxicity.

Acute porphyria

ROVANAEST solution for injection and infusion is possibly porphyrinogenic and should only be prescribed to patients with acute porphyria when no safer alternative is available. Appropriate precautions should be taken in the case of vulnerable patients, according to standard textbooks and/or in consultation with disease area experts.

Chondrolysis

There have been post-marketing reports of chondrolysis in patients receiving post-operative intra-articular continuous infusion of local anaesthetics, including Ropivacaine. The majority of reported cases of chondrolysis have involved the shoulder joint. Intra-articular continuous infusion is not an approved indication for ROVANAEST. Intra-articular continuous infusion with ROVANAEST should be avoided, as the efficacy and safety has not been established.

Excipients with recognised action/effect

This medicinal product contains sodium. To be taken into consideration by patients on a controlled sodium diet.

Prolonged administration

Prolonged administration of Ropivacaine should be avoided in patients concomitantly treated with strong CYP1A2 inhibitors, such as fluvoxamine and enoxacin.

Paediatric population

Neonates may need special attention due to immaturity of some organs and functions. This is especially important during continuous epidural infusion. Higher concentrations than 5 mg/mL have not been documented in children.

Interactions with Other Medicaments:

ROVANAEST should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics, e.g. certain antiarrhythmics, such as Lidocaine and Mexiletine, since the systemic toxic effects are additive. Simultaneous use of ROVANAEST with general anaesthetics or opioids may potentiate each other’s (adverse) effects. Specific interaction studies with Ropivacaine and anti-arrhythmic drugs class III (e.g. amiodarone) have not been performed, but caution is advised.

Cytochrome P450 (CYP) 1A2 is involved in the formation of 3-hydroxy-ropivacaine, the major metabolite. In vivo, the plasma clearance of Ropivacaine was reduced by up to 77% during co-administration of fluvoxamine, a selective and potent CYP1A2 inhibitor. Thus, strong inhibitors of CYP1A2, such as fluvoxamine and enoxacin given concomitantly during prolonged administration of ROVANAEST, can interact with ROVANAEST. Prolonged administration of Ropivacaine should be avoided in patients concomitantly treated with strong CYP1A2 inhibitors.

In vivo, the plasma clearance of Ropivacaine was reduced by 15% during co-administration of ketoconazole, a selective and potent inhibitor of CYP3A4. However, the inhibition of this isozyme is not likely to have clinical relevance.

In vitro, Ropivacaine is a competitive inhibitor of CYP2D6 but does not seem to inhibit this isozyme at clinically attained plasma concentrations.

Pregnancy and Lactation:

Pregnancy

Apart from epidural administration for obstetrical use, there are no adequate data on the use of Ropivacaine in human pregnancy. Experimental animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development.

Breastfeeding

There are no data available concerning the excretion of Ropivacaine into human milk.

Adverse Reactions:

The adverse reaction profile for ROVANAEST is similar to those for other amide local anaesthetics. Adverse reactions caused by the drug *per se* are difficult to distinguish from physiological effects of the nerve block (e.g. hypotension and bradycardia), and events caused directly (e.g. nerve trauma) or indirectly (e.g. epidural abscess) from needle puncture.

Undesirable effects (from all kinds of blockades):

The adverse reactions are displayed within each organ class with the following frequencies:

Very common (≥1/10), Common (≥1/100 to <1/10), Uncommon (≥1/1 000 to <1/100), Rare (<1/1 000)

Adults:

Organ System	Frequency	Adverse Reaction
Psychiatric disorders	Uncommon	Anxiety
Cardiac disorders	Common	Bradycardia ^a , Tachycardia, Hypertension
	Rare	Cardiac arrest, arrhythmias
Vascular disorders	Very Common	Hypotension ^a
	Common	Hypotension
	Uncommon	Syncope
Nervous System disorders	Common	Paraesthesia, Dizziness, Headache ^a
	Uncommon	Symptoms of CNS toxicity (Convulsions, Grand mal-convulsions, Seizures, Light-headedness, Circumoral paraesthesia, Numbness of the tongue, Hyperacusis, Tinnitus, Visual disturbances, Dysarthria, Muscular twitching, Tremor) ^b , Hypoaesthesia ^a
Respiratory, Thoracic and Mediastinal disorders	Uncommon	Dyspnoea ^{a*}
Gastrointestinal disorders	Very Common	Nausea
	Common	Vomiting ^{a,d}
Renal and Urinary disorders	Common	Urinary retention ^a
General disorders and Administrative site conditions	Common	Temperature elevation, Rigor, Back pain
	Uncommon	Hypothermia ^{a*}
	Rare	Allergic reactions (Anaphylactic reactions, angioneurotic oedema and urticaria)

^a These reactions are more frequent after spinal anaesthesia.

^b These symptoms usually occur because of inadvertent intravascular injection, overdose or rapid absorption.

^c Hypotension is uncommon in children (>1/100).

^d Vomiting is very common in children (>1/10).

Class-related adverse drug reactions

The adverse reactions below include complications related to the anaesthetic technique regardless of the local anaesthetic used.

Neurological complications

Neuropathy and spinal cord dysfunctions (e.g. anterior spinal artery syndrome, arachnoiditis, cauda equina) have been associated with intrathecal and epidural anaesthesia.

Total spinal block

Total spinal block may occur if an epidural dose is inadvertently administered intrathecally, or if a too large intrathecal dose is administered.

Acute systemic toxicity

Systemic toxic reactions primarily involve the central nervous system (CNS) and the cardiovascular system (CVS). Such reactions are caused by high blood concentration of local anaesthetics, which may appear due to (accidental) intravascular injection, overdose or exceptionally rapid absorption from highly vascularised areas. CNS reactions are similar for all amide local anaesthetics, while cardiac reactions are more dependent on the drug, both quantitatively and qualitatively.

Central nervous system toxicity

CNS toxicity is a graded response with symptoms and signs of escalating severity. The first symptoms are usually light-headedness, circumoral paraesthesia, numbness of the tongue, hyperacusis, tinnitus and visual disturbances. Dysarthria, muscular twitching and or tremors are more serious and precede the onset of generalized convulsions. These signs should not be mistaken for neurotic behaviour. Unconsciousness and grand mal convulsions may follow which may last for a few seconds to several minutes. Hypoxia and hypercarbia occur rapidly following convulsions due to increased muscular, together with the interference with respiration and possible loss of functional airways. In severe cases even apnoea may occur. Acidosis, hyperkalemia, hypocalcemia and hypoxia increase and extend the toxic effects of local anaesthetics.

Recovery is due to redistribution of the local anaesthetic drug from the central nervous system and subsequent metabolism and excretion. Recovery may be rapid unless large amounts of the drug have been injected.

Cardiovascular system toxicity

Cardiovascular system toxicity may be seen in severe cases and is generally preceded by signs of toxicity in the central nervous system. In patients under heavy sedation or receiving a general anaesthetic, prodromal CNS symptoms may be absent. Hypotension, bradycardia, arrhythmia and also cardiac arrest may occur as a result of high systemic concentrations of local anaesthetics but in rare cases, cardiac arrest has occurred without prodromal CNS effects.

In children, early signs of local anaesthetic toxicity may be difficult to detect since they may not be able to verbally express them, or if they are under general anaesthesia.

Symptoms and Treatment of Overdose:

Symptoms

Accidental intravascular injections of local anaesthetics may cause immediate (within seconds to a few minutes) systemic toxic reactions. In the event of overdose, peak plasma concentrations may not be reached for one to two hours, depending on the site of the injection, and signs of toxicity may thus be delayed.

Treatment

If signs of acute systemic toxicity appear, injection of the local anaesthetic should be stopped immediately and CNS symptoms (convulsions, CNS depression) must promptly be treated with appropriate airway/respiratory support and the administration of anticonvulsant drugs.

If circulatory arrest should occur, immediate cardiopulmonary resuscitation should be instituted. Optimal oxygenation and ventilation and circulatory support as well as treatment of acidosis are of vital importance.

If cardiovascular depression occurs (hypotension, bradycardia), appropriate treatment with intravenous fluids, vasopressor, and or inotropic agents should be considered. Children should be given doses commensurate with age and weight.

Should cardiac arrest occur, a successful outcome may require prolonged resuscitative efforts.

Effects on Ability to Drive and Use Machine:

Besides the direct anaesthetic effect, local anaesthetics may have a minor influence on mental function and co-ordination even in the absence of overt CNS toxicity and may temporarily impair locomotion and alertness.

Instructions for Use:

ROVANAEST solution for injection contains no preservative and is intended for single-use only. Remaining solution must be discarded.

The medicinal product should be visually inspected prior to use. The solution should only be used if it is clear, practically free from particles and if the container is undamaged.

Incompatibilities: Alkalisisation can cause precipitation, as Ropivacaine is hardly soluble at a pH above 6.0.

Storage condition: Do not store above 30°C.

Shelf Life:

3 years from manufacturing date in the proposed storage condition. Do not use after expiry.

Dosage Form and Packaging Available:

20 mL x 20 LDPE Ampoules per box

Manufacturer / Product Registration Holder:

 **AIN MEDICARE SDN.BHD.**
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

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